Does Music's Sensitive Period Leave a Trace in Episodic Memory's White Matter?

A Graph-Theoretic Test of the Chicken-and-Egg Problem Using HCP-D

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Abstract

Early music training is widely believed to help children's cognitive development, but it is still unclear whether the training itself is the real cause. Three main views appear in the literature: a *sensitive-period* account (when training starts matters), a *dose-response* account (how much one trains matters), and a *skeptical* account (apparent effects mostly come from other factors). I used HCP-D data ($N = 473$ with usable structural connectomes; trained-only $n \approx 183$) to test these views on episodic memory, measured by the Picture Sequence Memory (PSM) task. Music onset age and cumulative log-hours of training were put into a single regression model (M3) so that their effects could be compared on the same scale. Among the 11 graph features of the white-matter connectome (global efficiency, modularity, small-worldness, characteristic path length, and within/between-network strength for DMN, FPN, and MTL), only small-worldness was related to PSM at a nominal level ($\beta = +0.13$, raw $p = 0.019$), and this did not survive FDR correction. On the music side, the onset coefficient became more negative once cumulative hours was controlled (from $\beta = -0.092$ in M1 to $\beta = -0.207$ in M3), and it was somewhat larger in size than the hours coefficient ($\beta = -0.158$). The mediation analysis found no indirect effect for any of the 11 brain features (all 95% CIs included zero). Overall, the direction of the results leans toward the sensitive-period view, but neither coefficient reached $p < 0.05$. The lack of mediation also suggests that, if a sensitive-period effect exists in this sample, it does not seem to work through macroscale white-matter topology.

1 Background and Significance

My motivation for this project began with a common parental intuition: that early musical exposure makes children smarter. Many parents enroll their children in piano lessons before elementary school, hoping the experience will sharpen the brain. The popular literature reinforces this idea, but the empirical literature does not agree. Whether music training really improves general cognition has been debated for a long time, and the problem is a bit like a chicken-and-egg question.

Three positions occupy this debate. First, the **sensitive-period view** holds that the timing of training matters, that the developing brain has a window during which musical experience can sculpt neural circuitry in ways that mature brains cannot. Steele et al. (2013) provided the classic evidence: white-matter integrity in the posterior corpus callosum was systematically related to whether musicians began training before or after age seven, even after matching for total hours. Second, the **dose-response view** holds that the apparent onset effect is really an hours effect: children who start earlier accumulate more total practice by adulthood, and that cumulative dose is what matters, not the calendar age at first lesson (Schellenberg, 2020). Third, the **skeptical view** holds that neither variable carries information once one properly controls for the confounding effects of socioeconomic status, family environment, and pre-existing aptitude (Sala & Gobet, 2017).

These three positions cannot be told apart by a simple correlation. Because onset age and cumulative hours are negatively correlated, the relationship of either one with cognition is hard to interpret on its own. A more useful approach is a multiple regression that includes both predictors at the same time, because the coefficients on onset and hours, with the other held constant, show which view fits the data better. This is the main strategy I used in this project.

Episodic memory is a particularly appropriate behavioral outcome. Music learning is fundamentally sequential. Practice consists of encoding ordered temporal patterns, remembering motor sequences, and binding melodic events to contextual cues. The NIH Toolbox Picture Sequence Memory (PSM) task probes this same machinery: participants reproduce ordered sequences of pictured events. Compared with the broad construct of "intelligence," PSM offers a concrete, well-validated, theoretically motivated dependent variable that is available across all HCP-D participants. For this reason, PSM was selected as the primary cognitive outcome rather than as a proxy for general intelligence.

The white-matter side of the question is motivated by recent connectome-level work showing that structural network topology is itself developmentally dynamic. Mousley et al. (2025) reported four major lifespan turning points in white-matter topology (ages 9, 32, 66, and 83) and identified small-worldness as the strongest LASSO-retained predictor of age in the 9–32 epoch, the very epoch that contains the HCP-D age range. If music's sensitive period writes itself into the connectome, this is the developmental window in which we would expect that trace to be detectable.

This study looks at a meaningful question in developmental neuroscience: how early music training relates to episodic memory, and whether this shows up in the structure of brain connectivity. Even after many years of research on music and cognition, two problems are still not fully solved. The first is that onset age and cumulative training hours are naturally correlated, since children who start earlier usually also train more. This makes their effects on cognition easy to confuse. Studies that match participants on training hours can reduce some group differences, but they do not directly compare onset age and total practice as competing predictors. This project tries to fill that gap by estimating both effects in one model, which lets me check whether onset age explains memory beyond total training. The second problem is about the level of explanation. Most of the reported benefits of music training are at the behavioral level, and it is less clear whether they also appear in the structure of the connectome, especially for episodic memory. By connecting memory performance to white-matter network features through a mediation analysis, this study asks not only whether an association exists, but also at what level it might sit. The project also focuses on sequential episodic memory, which is closely related to music learning, instead of using broad measures of general intelligence. Beyond the theory, the question also has some practical value for early-childhood education, since it may help parents and teachers think about when and how much music training is worth investing in.

2 Research Questions and Hypotheses

2.1 Research questions

RQ1. Is music onset age associated with PSM performance? Does the association survive controlling for cumulative training hours?

RQ2. Is music onset age associated with white-matter network organization? If a behavioral effect exists, is it mediated by these network features?

2.2 Models

The first model tested the sensitive-period hypothesis by including only music onset age:

$$\text{M1}: \quad \text{PSM}_i = \beta_0 + \beta_1 \text{Onset}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{WISC}_i + \epsilon_i.$$

The second model tested the dose-response hypothesis by including only cumulative training hours:

$$\text{M2}: \quad \text{PSM}_i = \beta_0 + \beta_1 H_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \beta_4 \text{WISC}_i + \epsilon_i.$$

The third and primary model included both onset age and log-hours simultaneously:

$$\text{M3}: \quad \text{PSM}_i = \beta_0 + \beta_1 \text{Onset}_i + \beta_2 H_i + \beta_3 \text{Age}_i + \beta_4 \text{Sex}_i + \beta_5 \text{WISC}_i + \epsilon_i.$$

This model was central because onset age and cumulative training hours are naturally correlated: individuals who begin earlier often accumulate more total practice. By including both predictors in the same model, M3 tested whether earlier onset explained unique variance in PSM beyond training amount, and whether training amount explained unique variance beyond onset age.

The fourth model added an interaction term:

$$\text{M4}: \quad \text{PSM}_i = \beta_0 + \beta_1 \text{Onset}_i + \beta_2 H_i + \beta_3 \left(\text{Onset}_i \times H_i\right) + \beta_4 \text{Age}_i + \beta_5 \text{Sex}_i + \beta_6 \text{WISC}_i + \epsilon_i.$$

This interaction model tested whether the effect of starting age depended on the amount of training. The trained-only sample size for this step was approximately $n = 183$.

2.3 Pre-registered scenarios

The central behavioral question was whether PSM is better explained by the age at which music training began or by the cumulative amount of training. This comparison was tested in the primary joint model, M3, in which onset age and cumulative training hours were entered simultaneously:

$$\text{PSM} \sim \beta_1 \cdot \text{onset\_age} + \beta_2 \cdot \log(\text{hours}) + \text{covariates}$$

Here, a larger onset value means later music onset, whereas a larger hours value means greater cumulative training. Therefore, a sensitive-period account predicts a negative onset coefficient: later onset should be associated with lower PSM. By contrast, a pure dose-response account predicts a positive hours coefficient: greater cumulative training should be associated with higher PSM.

The three competing scenarios were therefore defined as follows:

- **H1a (Sensitive period):** $|\beta_1| > |\beta_2|$, with $\beta_1 < 0$. The timing of music onset should be the stronger predictor of PSM.

- **H1b (Dose-response):** $|\beta_2| > |\beta_1|$, with $\beta_2 > 0$. The cumulative amount of music training should be the stronger predictor of PSM.
- **H1c (Skeptical):** $\beta_1 \approx 0$, $\beta_2 \approx 0$, both coefficients are approximately zero after covariate adjustment. Any apparent music-training effect should disappear once confounders are controlled.

3 Methods

3.1 Step 1: Dataset and sample construction

Data come from the Human Connectome Project in Development (HCP-D). The starting demographic table contained 652 participants. Music variables were merged from the NDA SAIQ instrument (500 participants, 286 ever-trained). Cognitive variables (PSM, Flanker, DCCS, list sorting, pattern comparison) were merged from the corresponding NIH Toolbox files (611 participants with at least one cognitive score). After requiring a non-missing PSM score, 473 participants remained, all of which had usable structural connectivity matrices after quality control.

3.2 Step 2: Music variables

Music onset age was estimated as the participant's current age minus their reported total years of training. Participants reporting no training were treated as untrained and excluded from the trained-only analyses. Cumulative hours was estimated from the reported years × weeks/year × days/week × minutes/session, then log-transformed to compress the heavy right tail and stabilize variance.

3.3 Step 3: Structural connectomes and graph metrics

For each participant, the structural connectome was represented as a weighted undirected matrix

$$S \in \mathbb{R}^{246\times 246},$$

reconstructed from HCP-D diffusion MRI using DSI Studio's QSDR pipeline on the Brainnetome atlas. The raw edge weights corresponded to streamline counts. Before graph construction, each matrix was symmetrized and its diagonal was removed:

$$W_{ij}^{(0)} = \begin{cases} \frac{1}{2}\left(S_{ij} + S_{ji}\right), & i \neq j, \\ 0, & i = j. \end{cases}$$

Because streamline counts were highly right-skewed, edge weights were stabilized with a log transform:

$$W_{ij}^{(\log)} = \log\left(1 + \max\left(W_{ij}^{(0)}, 0\right)\right).$$

For the global graph-theoretic metrics, matrices were proportionally thresholded. For a density level ρ, the number of retained undirected edges was

$$K_\rho = \operatorname{round}\left(\rho \frac{N(N-1)}{2}\right),$$

where $N = 246$. The set E_ρ contained the K_ρ strongest off-diagonal edges in the upper triangle of $W^{(\log)}$. The thresholded matrix was therefore

$$W_{ij}^{(\rho)} = \begin{cases} W_{ij}^{(\log)}, & \{i, j\} \in E_\rho, \\ 0, & \text{otherwise.} \end{cases}$$

The primary density was $\rho = 0.15$, with robustness checks at $\rho = 0.10$ and $\rho = 0.20$. After thresholding, weights were normalized by the maximum retained weight:

$$A_{ij}^{(\rho)} = \frac{W_{ij}^{(\rho)}}{\max_{u,v} W_{uv}^{(\rho)}},$$

so that all nonzero edge weights lay in $[0, 1]$. All subsequent global graph metrics were computed on this weighted undirected matrix $A^{(\rho)}$.

Node strength was defined as the weighted degree

$$s_i = \sum_{j=1}^{N} A_{ij},$$

and total strength was computed as the sum of all upper-triangular edge weights:

$$S_{\text{total}} = \sum_{i<j} W_{ij}^{(\log)}.$$

Global efficiency was computed as the average inverse shortest-path distance between all node pairs:

$$E_{\text{glob}} = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}},$$

where d_{ij} is the length of the shortest weighted path between nodes i and j. To convert connection weights into path lengths, stronger connections were treated as shorter distances:

$$\ell_{ij} = \begin{cases} 1/A_{ij}, & A_{ij} > 0, \\ \infty, & A_{ij} = 0. \end{cases}$$

The shortest-path distance was therefore

$$d_{ij} = \min_{p: i \to j} \sum_{(u,v) \in p} \ell_{uv}.$$

Characteristic path length was computed from the same shortest-path distance matrix:

$$L = \frac{1}{N(N-1)} \sum_{i \neq j} d_{ij},$$

excluding disconnected node pairs when distances were infinite. Thus, both global efficiency and characteristic path length reflect a shortest-path communication model, where information is assumed to travel along the strongest and shortest available routes in the structural network.

The weighted clustering coefficient was computed at the node level and then averaged across nodes:

$$C = \frac{1}{N} \sum_{i=1}^{N} C_i.$$

For each node i, C_i quantified the tendency of its neighbors to also be connected to each other, weighted by the geometric mean of triangle edge weights:

$$C_i = \frac{\sum_{j,k} (A_{ij} A_{ik} A_{jk})^{1/3}}{k_i(k_i - 1)},$$

where k_i is the number of nonzero neighbors of node i. This measure is high when a node participates in strongly weighted local triangles.

Modularity was computed by partitioning the graph into communities and maximizing the quality function

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{s_i s_j}{2m} \right] \delta(g_i, g_j),$$

where

$$m = \frac{1}{2} \sum_{i,j} A_{ij},$$

s_i and s_j are node strengths, g_i and g_j are the community labels of nodes i and j, and $\delta(g_i, g_j) = 1$ if the two nodes belong to the same community and 0 otherwise. Larger Q indicates stronger separation into within-community connections relative to what would be expected from node strength alone.

Small-worldness was computed by comparing the observed network's clustering and path length to degree-preserving randomized null networks:

$$\sigma = \frac{C/C_{\text{rand}}}{L/L_{\text{rand}}},$$

where C and L are the observed mean clustering coefficient and characteristic path length, and C_{rand} and L_{rand} are their averages across randomized networks. The randomized networks preserved the degree sequence while rewiring edges, providing a null model for the expected clustering and path length of a network with comparable degree structure. A value $\sigma > 1$ indicates a network with greater local clustering than a comparable random graph while maintaining relatively short path lengths.

In addition to global graph metrics, hypothesis-driven network-level connection strengths were computed for three Yeo-defined subnetworks: the default mode network (DMN), frontoparietal network (FPN), and medial temporal/limbic network (MTL). Unlike the global graph metrics, these within- and between-network features were computed from the full log-transformed SC matrix rather than the proportionally thresholded matrix. To remove participant-level multiplicative scaling effects, the full log-transformed matrix was normalized by its maximum edge weight:

$$\widetilde{W}_{ij} = \frac{W_{ij}^{(\log)}}{\max_{u,v} W_{uv}^{(\log)}}.$$

For a network R with node set V_R, within-network strength was defined as the mean normalized edge weight among all pairs of nodes inside that network:

$$\mathrm{Within}(R) = \frac{2}{|V_R|(|V_R|-1)} \sum_{\substack{i<j \\ i,j \in V_R}} \widetilde{W}_{ij}.$$

For two distinct networks R and T, between-network strength was defined as

$$\mathrm{Between}(R,T) = \frac{1}{|V_R||V_T|} \sum_{i \in V_R} \sum_{j \in V_T} \widetilde{W}_{ij}.$$

This yielded six network features:

$$\mathrm{DMN}_{\mathrm{within}}, \quad \mathrm{FPN}_{\mathrm{within}}, \quad \mathrm{MTL}_{\mathrm{within}},$$

$$\text{DMN-FPN}_{\mathrm{between}}, \quad \text{DMN-MTL}_{\mathrm{between}}, \quad \text{FPN-MTL}_{\mathrm{between}}.$$

Together, the graph metrics captured global integration, local segregation, community structure, and small-world organization, while the network-level features captured anatomically motivated connection strengths among memory- and control-related subnetworks. More decentralized communication models, such as greedy navigation, random walks, diffusion, or communicability, were not modeled in the present analysis.

3.4 Statistical models

All confirmatory analyses were conducted using multiple linear regression. Across all models, age, sex, and WISC-V Matrix Reasoning were included as covariates to control for developmental, demographic, and general fluid-reasoning differences. Continuous predictors were z-standardized before model fitting:

$$X_z = \frac{X - \bar{X}}{\mathrm{SD}(X)}.$$

This allowed regression coefficients to be interpreted on a comparable scale: each standardized coefficient β represents the expected change in the outcome, in standard-deviation units, for a one-standard-deviation increase in the predictor, holding covariates constant.

Because multiple statistical tests were performed within each analysis step, false discovery rate correction was applied using the Benjamini–Hochberg procedure. Correction was performed separately within each step: Step 4 corrected across the 11 brain graph features, Step 5 corrected across the two primary predictors in the joint music model, Step 6 corrected across 22 music–brain tests, and Step 7 corrected across the 11 mediation tests.

Let

$$H_i = \log(1 + \mathrm{Hours}_i)$$

denote the log-transformed cumulative music-training hours for participant i.

Step 4: Brain → PSM. First, we tested whether individual differences in structural connectome organization predicted episodic memory performance. Picture Sequence Memory (PSM) score was used as the outcome variable, and each of the 11 graph-theoretic features was entered as a predictor in a separate regression model:

$$\mathrm{PSM}_i = \beta_0 + \beta_1 G_{ik} + \beta_2 \mathrm{Age}_i + \beta_3 \mathrm{Sex}_i + \beta_4 \mathrm{WISC}_i + \epsilon_i,$$

where G_{ik} denotes graph feature k for participant i. This analysis tested whether global efficiency, modularity, clustering coefficient, characteristic path length, small-worldness, or network-specific connection strengths were associated with PSM after controlling for covariates. The available sample size for this brain–behavior analysis was $n = 306$.

Step 5: Music → PSM. Second, we tested whether music-training history predicted episodic memory performance among participants with music-training experience. This trained-only subsample was used because onset age and cumulative training hours are meaningful only for participants who had received music training. The four nested regression models described above were used.

Step 6: Music → Brain. Third, we tested whether music-training variables were associated with structural connectome organization. Each of the 11 graph features was used as an outcome variable. Onset age and log-hours were entered jointly into a single covariate-adjusted model, so that each variable's partial effect was estimated holding the other constant:

$$G_{ik} = \beta_0 + \beta_1 \mathrm{Onset}_i + \beta_2 H_i + \beta_3 \mathrm{Age}_i + \beta_4 \mathrm{Sex}_i + \beta_5 \mathrm{WISC}_i + \epsilon_{ik}.$$

This step asked whether earlier music onset or greater cumulative training was reflected in white-matter network topology. Because each of the 11 graph features yielded a partial coefficient for both onset age and log-hours, this step produced 22 music–brain partial-effect tests. The available sample size for this analysis was $n = 175$.

Step 7: Mediation analysis. Finally, we tested whether the association between music onset age and PSM could be statistically explained by structural connectome features. For each graph feature G_k, a mediation model was estimated with onset age as the predictor, the graph feature as the mediator, and PSM as the outcome. Log cumulative hours H_i was included as a covariate in both paths, so the mediation estimates the onset effect holding cumulative training constant. The first path estimated whether onset age predicted the brain feature:

$$G_{ik} = \alpha_0 + a\,\mathrm{Onset}_i + \alpha_1 \mathrm{Age}_i + \alpha_2 \mathrm{Sex}_i + \alpha_3 \mathrm{WISC}_i + \alpha_4 H_i + \epsilon_{ik}^{(a)}.$$

The second path estimated whether the brain feature predicted PSM after controlling for onset age and cumulative hours:

$$\mathrm{PSM}_i = \gamma_0 + c'\,\mathrm{Onset}_i + b\,G_{ik} + \gamma_1 \mathrm{Age}_i + \gamma_2 \mathrm{Sex}_i + \gamma_3 \mathrm{WISC}_i + \gamma_4 H_i + \epsilon_i^{(b)}.$$

The indirect effect was defined as

$$a \times b,$$

which quantifies the extent to which onset age is associated with PSM through its association with a given graph feature. A 95% bootstrap confidence interval was estimated using 5,000 resamples. This mediation step therefore tested the pathway

$$\text{Music onset} \rightarrow \text{Brain graph feature} \rightarrow \text{PSM}.$$

The available sample size for mediation was $n = 175$.

Overall, the statistical analysis proceeded in four linked stages. Step 4 tested whether brain network features predicted episodic memory. Step 5 tested whether music-training history predicted episodic memory. Step 6 tested whether music-training history predicted brain network features. Step 7 tested whether brain network features mediated the relationship between music onset age and episodic memory. Together, these models evaluated not only whether music training was associated with PSM, but also whether such an association could be explained at the level of structural connectome organization.

4 Results

4.1 Sample description

The trained subsample ($n = 201$ with complete music variables) has an onset age distribution heavily concentrated between 8 and 11 years (Fig. 1). Only 44 participants ($\approx 22\%$) started before age 7. This distribution shapes the interpretive boundary of the project: we are testing "relatively earlier vs. later" onset within a typical-population sample, not the classical pre-7 versus post-7 sensitive-period contrast that motivated Steele et al. (2013).

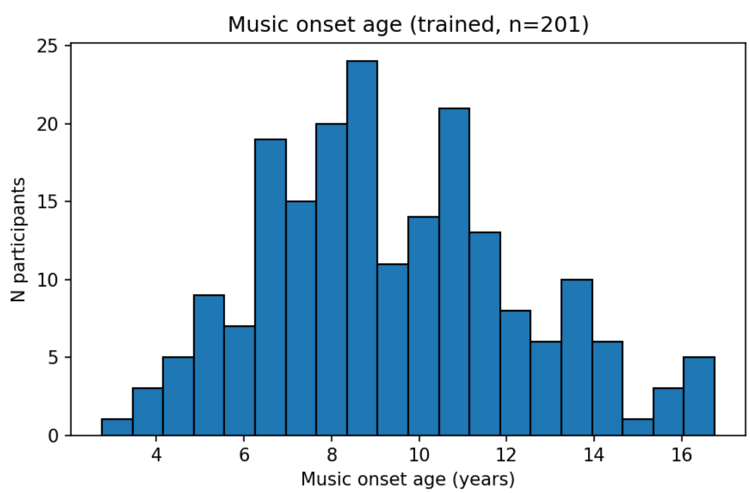

Figure 1: Distribution of estimated music onset age in the trained subsample ($n = 201$).

4.2 Brain → PSM (Step 4)

Figure 2 shows the standardized coefficients from the 11 covariate-adjusted regressions that tested whether each white-matter graph feature predicted PSM. Among the 11 features, small-worldness had the largest positive coefficient with PSM ($\beta = +0.130$, raw $p = 0.019$), followed by modularity. Several features were close to zero, while FPN within-network strength and characteristic path length were negative. None of these survived FDR correction (all q-values above 0.21), so Step 4 gives only a weak, nominal brain–behavior signal. The positive direction of small-worldness still fits the idea that better episodic memory may go with a more balanced integration–segregation profile in the developing connectome.

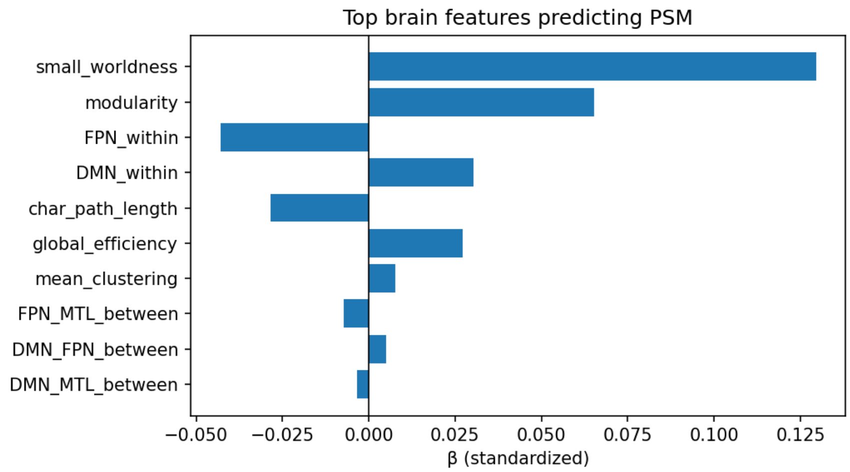

Figure 2: Standardized regression coefficients for each graph-theoretic feature predicting PSM in the Step 4 Brain → PSM analysis ($n = 306$). Positive values indicate higher PSM with larger graph-feature values, whereas negative values indicate lower PSM. Small-worldness showed the largest nominal positive association, but all FDR-corrected q-values exceeded 0.21.

4.3 Music → PSM (Step 5)

Before looking at the adjusted models, the raw group pattern already shows why a joint model is needed. In the unadjusted boxplot, the late-onset group has a slightly higher median PSM score than the early- or middle-onset groups (Fig. 3). Taken at face value, this descriptive pattern appears inconsistent with a sensitive-period account. However, the raw comparison does not adjust for developmental and training-history differences. In HCP-D, later starters tend to be older, and older participants have more developmentally mature PSM performance. The raw plot therefore mixes onset timing with age and cumulative training history.

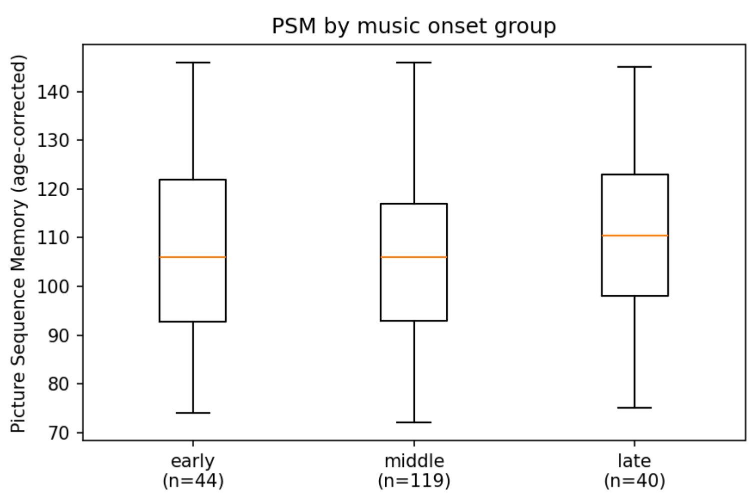

Figure 3: Raw, unadjusted PSM by music onset group. The late-onset group has a marginally higher median than the early-onset group, a pattern that reverses once cumulative hours and demographic covariates are controlled.

The adjusted models explain this reversal. When entered alone, neither onset age (M1: $\beta = -0.092$, $p = 0.30$) nor cumulative hours (M2: $\beta = -0.039$, $p = 0.63$) was significant. In the joint model M3, onset age became more negative ($\beta = -0.207$, $p = 0.073$), while cumulative hours stayed smaller in size ($\beta = -0.158$, $p = 0.129$). So once the shared variance with hours was removed, the onset coefficient grew clearly larger. The onset $\times$ hours interaction in M4 was about zero ($\beta = +0.030$, $p = 0.742$), which means the two predictors mostly add up rather than depending on each other.

The main result of the project is the comparison of M1, M2, and M3 (Table 1, Fig. 4). In M3, the onset coefficient was larger in size than the hours coefficient, with a ratio of about $0.207/0.158 \approx 1.31$.

Table 1: Standardized coefficients across nested music–PSM models. The trained-only subsample was used ($n \approx 183$). M3 is the pre-registered primary model.

Model	Predictor	β	p
M1 (onset only)	onset age	-0.092	0.300
M2 (hours only)	log hours	-0.039	0.629
M3 (both)	**onset age**	$\mathbf{-0.207}$	0.073
M3 (both)	**log hours**	$\mathbf{-0.158}$	0.129
M4 (+ interaction)	onset age	-0.215	0.070
M4 (+ interaction)	log hours	-0.160	0.126
M4 (+ interaction)	onset $\times$ hours	$+0.030$	0.742

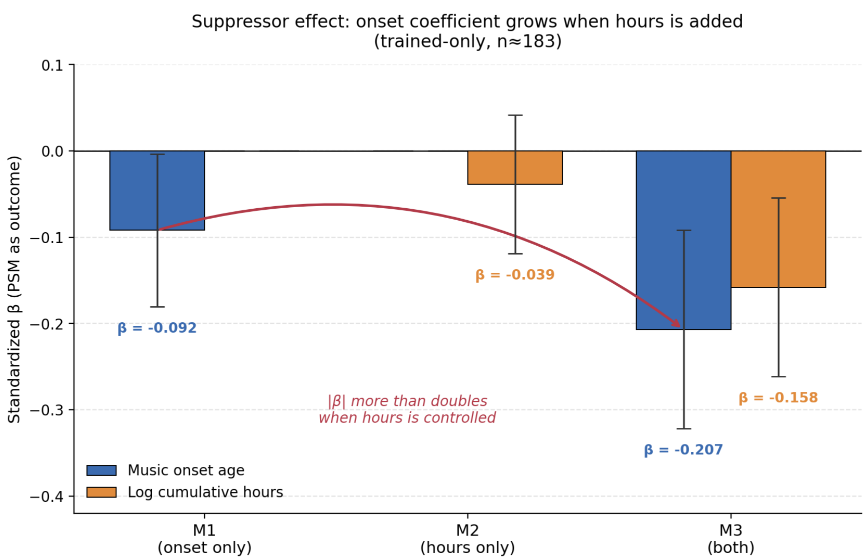

Figure 4: Suppressor pattern across the M1, M2, M3 nested model sequence. Error bars are ± 1 SE. The curve traces the growth in onset's coefficient from -0.092 in M1, the marginal model, to -0.207 in M3, the partial model.

This pattern is informative. A pure dose-response account would expect the hours coefficient to be the larger one once both predictors are included, and to be positive. The data show the opposite: the onset coefficient was larger than the hours coefficient (the ratio above, ≈ 1.31). The negative onset coefficient fits the sensitive-period idea that later onset goes with lower PSM. The negative hours coefficient does not fit a positive dose-response story. The near-zero interaction also suggests the onset effect does not depend on reaching some amount of training.

In M3, onset age and log cumulative hours were only weakly correlated ($r = -0.27$), and their variance inflation factors were 2.37 and 2.14, both well below the usual cutoff of 5. So the change in the onset coefficient comes from separating the two predictors' effects, not from multicollinearity making the estimates unstable.

Overall, the M3 pattern points in the direction of H1a: onset age had the larger coefficient and it was negative. But since neither coefficient reached significance, I read this as weak, exploratory support for an onset-dominant pattern, not as firm evidence for a sensitive period.

4.3.1 Secondary cognitive outcomes: specificity of the effect

To test whether the PSM effect is specific to episodic memory or instead reflects a broader effect across general cognition, the same nested models were repeated for four secondary cognitive outcomes: Flanker (inhibitory control), DCCS (cognitive flexibility / set-shifting), List Sorting Working Memory, and Pattern Comparison Processing Speed. These analyses were conducted in the trained-only subsample ($n = 183$). Table 2 summarizes the standardized partial coefficients of onset age and log cumulative hours from the primary joint model M3 for each outcome.

Table 2: Standardized partial regression coefficients from M3, in which onset age and log cumulative hours were entered jointly. The trained-only subsample was used ($n = 183$). PSM, the primary outcome, is included for comparison.

Outcome (domain)	onset β	hours β	onset p	$\lvert\beta_{\text{onset}}\rvert/\lvert\beta_{\text{hours}}\rvert$
PSM (episodic memory, reference)	$-$**0.207**	-0.158	0.073	1.31
DCCS (cognitive flexibility)	-0.232	-0.015	0.054	15.0
List Sorting (working memory)	-0.189	-0.141	0.119	1.34
Pattern Comparison (processing speed)	-0.122	$+0.054$	0.288	2.28
Flanker (inhibitory control)	-0.043	$+0.128$	0.720	0.34

No secondary outcome survived FDR correction, and the trained-only sample ($n = 183$) is small for a model with several predictors. So the points below are exploratory and describe the general pattern rather than a firm result. Even so, two things stand out. First, the onset coefficient was larger than the hours coefficient in three of the four secondary outcomes (DCCS, List Sorting, and Pattern Comparison). This suggests the onset-leaning pattern seen for PSM in Section 4.3 is not just a feature of one single outcome. The onset coefficient for DCCS in particular ($\beta = -0.232$, $p = 0.054$) was a little larger than for PSM ($\beta = -0.207$) and was close to significance.

Second, the size of the onset effect seemed to depend on the type of task. It was strongest for tasks that involve sequencing, memory, and executive control (PSM, DCCS, and List Sorting), and weaker or absent for simple processing speed and inhibition. Flanker was the only outcome where hours had a larger coefficient than onset. This pattern looks like near transfer (Bigand & Tillmann, 2022): the effect of music onset shows up mostly in abilities that are close to what music learning actually trains, such as sequencing and memory, and less in abilities that have little in common with music. This fits a selective sensitive-period idea better than a skeptical one, in which music training would just raise general ability across the board. Still, since none of the coefficients were FDR-significant and the trained-only sample was small, I treat this as an idea for future work rather than a conclusion.

4.4 Music → Brain (Step 6)

Eleven joint regressions were estimated, one per graph feature, each yielding a partial coefficient for onset age and for log cumulative hours (22 partial-effect tests in total). None survived FDR correction. The most prominent raw signals were DMN–FPN between-network connectivity with log cumulative hours ($\beta = +0.13$, $p = 0.25$) and small-worldness with onset age ($\beta = -0.13$, $p = 0.28$). Both effects were too weak to read much into on their own, but their directions matched what we expected: more practice went with stronger between-network connectivity, and later onset went with less small-world structure.

4.5 Mediation: Onset → Brain → PSM (Step 7)

Across all 11 candidate mediators, the bootstrap 95% confidence intervals for the indirect effect ($a \times b$) included zero (Fig. 5). The largest point estimate was observed for small-worldness ($a \times b = -0.016$, CI $[-0.063, +0.020]$, $p_{\text{boot}} = 0.40$). Its direction was consistent with the sensitive-period prediction, but the indirect effect was statistically null.

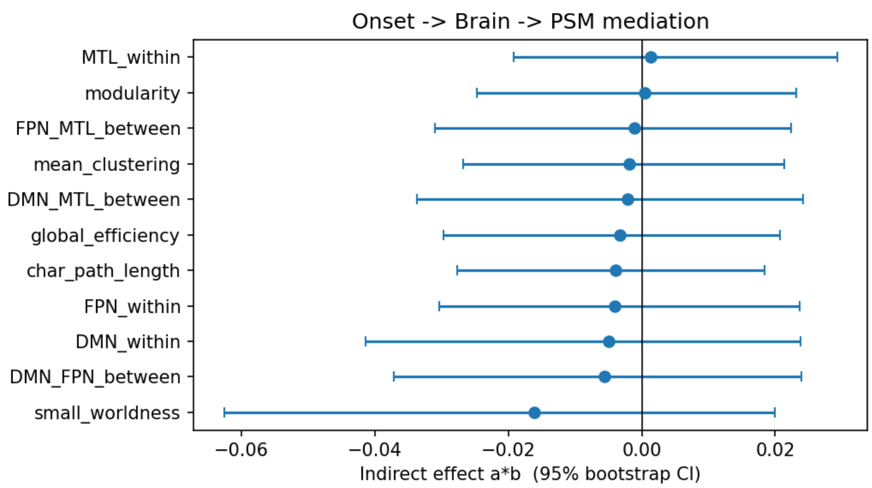

Figure 5: Indirect effects of music onset age on PSM through each of the 11 candidate white-matter mediators. Whiskers indicate 95% bootstrap confidence intervals based on 5,000 resamples. All intervals include zero.

One more thing to note: the total effect of onset age on PSM (the c-path, $\beta = -0.236$) barely changed when each mediator was added, dropping only to about $c' \approx -0.22$ in every model. This means the onset → PSM link does not really pass through the macroscale white-matter graph features tested here.

5 Discussion

5.1 Verdict on the three scenarios

The **sensitive-period scenario** fits best out of the three, but only weakly. The main support is M3: onset age had a larger coefficient than cumulative hours, and the onset coefficient was negative, which is what H1a predicts. In other words, later music onset went with lower PSM after hours and the covariates were controlled. Two things hold this back, though. First, neither coefficient reached $p < 0.05$, with the onset coefficient sitting around $p \approx 0.07$. Second, the mediation through brain features was basically absent. So if a sensitive-period effect is there in this sample, it does not seem to run through the 11 topology features I measured.

The **dose-response scenario** is weakly against the data. A pure dose-response account would expect cumulative hours to be the stronger predictor and to be positive. Instead, with onset held constant, the hours coefficient was smaller than the onset coefficient and negative. The interaction was also near zero, so there was no sign that more training amplified an onset effect.

The **skeptical scenario cannot be confidently rejected**. Nothing survived FDR correction, the trained-only sample ($n = 183$) is small for a model with several predictors, and, importantly, my covariates (age, sex, fluid reasoning) do not include socioeconomic status or family environment, which are exactly the confounders the skeptical view cares about. So I cannot rule out that the onset pattern partly reflects background variables I did not measure.

5.2 Why the macroscale connectome may be silent

Maybe the most useful negative result is the mediation null. Steele et al. (2013) found their sensitive-period effect in the posterior corpus callosum, using fractional anisotropy from tractography, which is a microstructural and tract-specific measure. My 11 features are instead network-level summaries of the whole-brain graph after thresholding. Two explanations seem possible:

- **Wrong spatial scale.** A localized FA difference in a specific commissural tract may not change network-level topology metrics like small-worldness or modularity. The signal could be there at the tract level and invisible at the global level.
- **Wrong tissue.** Sequence memory depends critically on hippocampal–prefrontal circuitry. Gray-matter volume in MTL or functional connectivity between MTL and FPN may be a more direct substrate than aggregate structural-connectome topology.

Either way, the better next step is probably not to just rerun the same pipeline with a larger N. It would be to redesign it with tract-specific microstructure (FA, RD) in tracts of interest, or with hippocampal/MTL volume, on the same HCP-D sample.

5.3 Connection to the lifespan turning-points framework

The one nominally significant Brain $\rightarrow$ PSM signal, small-worldness ($\beta = +0.13$), may not be random. Mousley et al. (2025) reported that small-worldness has the largest LASSO coefficient for predicting age in the 9–32 lifespan epoch, which is exactly the epoch that covers the HCP-D age range. The fact that my only behavioral hit is in the same metric that the lifespan paper highlights for this age window is at least consistent with the idea that PSM-related connectome changes in this period partly involve small-worldness. This is just a hypothesis that the present study cannot settle, but it is worth following up.

5.4 Limitations

- **Power.** Trained-only $n = 183$. The M3 onset coefficient sits at $p = 0.07$, in an awkward range where adding maybe 50–100 more participants could probably settle the question one way or the other.

- **Onset coverage.** Only $\approx$ 22% of trained participants started before age 7, so the classical Steele-style early-vs-late contrast is limited by the makeup of the sample, not just its size.
- **Unmeasured confounders.** No socioeconomic-status or family-income covariate was available in the analysis sample, so the skeptical scenario cannot be fully addressed.
- **Self-report.** Music history is retrospective parental/self-report, and cumulative hours is estimated from coarse multiplicative units that compound measurement noise.
- **Cross-sectional.** Causal direction cannot be pinned down, so "better brain → earlier start" is still formally indistinguishable from "earlier start → better brain."
- **Macroscale features only.** We did not test tract-specific FA/RD, gray-matter volume, or functional connectivity, any of which could carry the trace that macroscale topology does not.

6 Conclusion

This project asked whether the timing of music onset leaves a trace in episodic memory and in the macroscale white-matter connectome. Using the three-scenario framework, I found three things. For *behavior*, the onset effect on PSM grew from about $|\beta| = 0.09$ to $|\beta| = 0.21$ once cumulative hours was controlled, and it was larger than the hours effect ($p = 0.073$, $n = 183$), which is the pattern the sensitive-period view predicts. For the *brain*, only small-worldness predicted PSM at a nominal level (raw $p = 0.019$, not FDR-significant), and this is the same metric that lifespan work (Mousley et al., 2025) flags as the strongest age predictor for the 9–32 epoch that my sample falls in. For *mediation*, none of the macroscale topology features carried the onset → PSM signal (all 11 indirect-effect 95% CIs included zero). The honest summary is that the behavioral results lean toward the sensitive-period view, but the trace, if it is real, does not seem to live in macroscale white-matter topology. The most useful next step would be to look instead at tract-specific microstructure and at MTL gray-matter and functional measures, which is where the original sensitive-period and episodic-memory literatures point.

7 Post-Presentation Reflection and Future Directions

This section was written after the presentation, incorporating questions and discussion from peers.

7.1 Comparison with measures beyond PSM

Honestly, if IQ had been available in this dataset, I would have included it, since IQ is the first thing that comes to mind when you think about "intelligence" in the most general sense. Still, beyond PSM, I did get some extra results using four more measures (Flanker, DCCS, LSWMT, PCPS).

7.2 All predictors such as onset age and cumulative hours were z-scored

Because these variables are on different absolute scales, they were entered into the models as standardized (z-scored) values, so that β_{onset} and β_{hours} in M3 could be directly compared. Cumulative hours was log-transformed before z-scoring, since its distribution was skewed.

7.3 Personal interpretation

Although neither onset nor hours was significant in M3, the direction of the partial regression coefficients was consistent with the sensitive-period account ($|\beta_{\text{onset}}| > |\beta_{\text{hours}}|$). Given the cross-sectional design, the absence of an SES covariate, and the modest sample size, these data alone cannot settle the chicken-or-egg question. That said, my own intuition is that the sensitive-period hypothesis (H1a) still seems plausible, and I think a much larger and more precise longitudinal dataset, with detailed training histories, could show it more clearly. My plan to have my (future) children learn music is something I have held for a very long time; even if pursuing my own bucket list through raising a child is admittedly a bit selfish, I would still want to do it if I can.

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